

CHM111: General Chemistry I: Practice Exam

Dr. Deskins

Name: _____

This practice exam covers Chapters 1, 2 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Identify the state of matter in which iron exists at 25 °C and 1 atm.

ANSWER: _____

2. Classify the following as a physical or chemical change: Burning propane in a gas grill. Explain your answer.

ANSWER: _____

3. An engineer measures the capacity of an electricity-generating plant. The capacity is 879 MW. What is the capacity in watts? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

4. Provide the element symbol that represents nickel.

ANSWER: _____

5. Write the chemical formula for chromic acid.

ANSWER: _____

6. Classify the following as an element or a compound: pure sulfur dioxide gas. You must explain your answer using correct reasoning.

ANSWER: _____

7. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$400 \text{ mol} \div 28. \text{ L}$$

ANSWER: _____

8. A botanist measures the thickness of a leaf. The thickness is 4.3 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

9. Does the following process produce a new substance? A glass of water is left uncovered in a warm room and the water level slowly drops. You must explain your answer with valid reasoning.

ANSWER: _____

10. A millworker measures the width of a plank. The width is 7.7 dm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

11. Provide the chemical formula for the molecular compound silicon tetrafluoride.

ANSWER: _____

12. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 7.08×10^{10} m. What is the distance in gigameters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

13. Katherine looks up the density of a material and finds it listed as $3.089 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

14. Classify the following as a homogeneous or heterogeneous mixture: muddy river water that contains visible suspended particles. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

15. Provide the element symbol that represents bromine.

ANSWER: _____

16. A shipwright measures the length of a bolt. The length is 57.9 cm. What is the length in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

17. A lab technician measures the weight of a reagent. The weight is 3284 g. What is the weight in kilograms? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

18. Nicholas measures the mass of a glass marble using a balance. The accepted mass of the object is 22.8 g. He records a measurement of 22.02 g. What is the percent error of his measurement? Report your answer to two significant figures.

ANSWER: _____

19. Patricia needs to determine the mass of a sample of acetone. The density of the sample is $1.65 \frac{\text{g}}{\text{mL}}$ and its volume is 10.9 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

20. An optician measures the diameter of her patient's pupil. The diameter is 0.0016 m. What is the diameter in millimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

21. Christopher needs to determine the mass of a sample of ethanol. The density of the sample is $16.32 \frac{\text{g}}{\text{mL}}$ and its volume is 28.83 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

22. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 64 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

23. A nuclear chemist measures the mass of an artificial element. The mass is 28 pg. What is the mass in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

24. Julia measures the speed of a moving object and records $18.62 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

25. A pharmacist measures the mass of a powdered drug. The mass is 9383 mg. What is the mass in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

26. A nurse in a U.S. hospital records a patient's temperature as 41 °F. Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

27. Identify the state of matter in which iron exists at 25 °C and 1 atm.

ANSWER: _____

28. An optician measures the diameter of her patient's pupil. The diameter is 0.0087 m. What is the diameter in millimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

29. Classify color as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

30. As part of his duties as an analytical chemist, Nathan analyzes a sample of an unknown pure substance X and finds that it contains 6.8 g of element E₁ and 12.8 g of element E₂, and no other elements. He then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in his lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
X	6.8 g	12.8 g
A	26.1 g	8.2 g
B	5.3 g	5.0 g
C	28.7 g	18.4 g

ANSWER: _____

31. Jonathan measures the density of a saline solution in his lab. He finds that the sample has a mass of 7.0 g and a volume of 14.75 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

32. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 8.54×10^{-14} m. What is the wavelength in femtometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

33. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 24 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

34. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 7.1828×10^{-7} m. What is the wavelength in nanometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

35. Write the name of the inorganic acid H₃AsO₄.

ANSWER: _____

36. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 2.6970×10^{-7} m. What is the wavelength in nanometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

37. A cell biologist measures the width of a cell. The width is $66\ \mu\text{m}$. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

38. Provide the chemical symbol of the most commonly formed ion of barium and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

39. Classify F_2O as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

40. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$6.943\text{ cm} + 131.14\text{ cm}$$

ANSWER: _____

41. Classify the ability to rust of iron as a physical property or a chemical property. Explain your answer.

ANSWER: _____

42. A civil engineer measures the length of an airport runway. The length is 8549 m. What is the length in kilometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

43. Nicholas is comparing two unknown liquid samples. Sample A has a mass of 38 g and a volume of 42 mL. Sample B has a mass of 75 g and a volume of 27.8 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

44. Classify Cl_4Si as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

45. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.280 m. What is the thickness in centimeters? Report your answer with units. Report your answer in scientific notation.

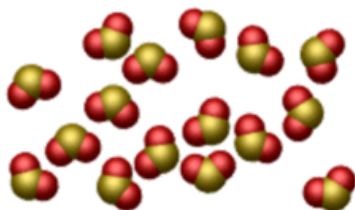
ANSWER: _____

46. As part of her duties as an analytical chemist, Abigail measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Abigail would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	6.4 g	4.0 g
2	18.2 g	11.4 g
3	9.3 g	5.8 g
4	27.7 g	17.3 g

ANSWER: _____

47. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

48. Classify the following as a pure substance or a mixture: a glass of orange juice with pulp. You must explain your answer and include definitions.

ANSWER: _____

49. Sean needs to determine the mass of a sample of ethanol. The density of the sample is $1.29 \frac{\text{g}}{\text{mL}}$ and its volume is 29.83 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

50. Write the name of the inorganic acid H_2SO_4 .

ANSWER: _____

51. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: air at sea level, which appears clear and uniform throughout. You must explain your answer with definitions.

ANSWER: _____

52. Provide the name and empirical formula of an ionic compound that is formed by the silver cation and the chloride anion.

ANSWER: _____

53. Classify the following as a pure substance or a mixture: a handful of garden soil. You must explain your answer and include definitions.

ANSWER: _____

54. Determine whether chlorine and iodine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

55. Provide the number of protons and the number of neutrons there are in an atom of ^{51}V .

ANSWER: _____

56. Mary is a chemical engineer who needs to convert an oven temperature for a thermodynamic calculation. The temperature is 23°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

57. Determine whether chromium and iodine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

58. Classify the following as a homogeneous or heterogeneous mixture: a solution of table sugar completely dissolved in water. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

59. A nurse records a patient's temperature. The thermometer reads 91.40°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

60. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$2.8436\text{ mL} - 0.23\text{ mL}$$

ANSWER: _____

61. A physical chemist measures the width of a molecule. The width is 47 nm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

62. Ethan needs to find the volume of a 36.43 g sample of glycerin. The density of the sample is $0.7725 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

63. Classify the following as a homogeneous or heterogeneous mixture: a mixture of oil and water that has separated into two distinct layers. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

64. A civil engineer measures the length of an airport runway. The length is 8362 m. What is the length in kilometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

65. Sam needs to find the volume of a 9.221 g sample of a concentrated solution. The density of the sample is $3.1 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

66. Samantha needs to know the mass of a sample of BaBr_2 to 4 significant figures. She puts the sample on a digital scale. This is what the scale shows: 90.4568 g. If this measurement is precise enough for Samantha, round it to 4 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

67. Provide the name for the molecular compound ClF. If the compound has a common name, you may provide that instead.

ANSWER: _____

68. A shipwright measures the length of a bolt. The length is 65.3 cm. What is the length in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

69. Provide the name of the element that the element symbol Hg represents.

ANSWER: _____

70. Provide the name and empirical formula of the ionic compound that is formed by the Ba^{2+} cation and the S^{2-} anion.

ANSWER: _____

71. Classify the following as a physical or chemical change: Freezing liquid water into ice cubes. Explain your answer.

ANSWER: _____

72. A recipe instructs a cook to preheat an oven to 122 °F. Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

73. Write the name of the inorganic acid HClO_3 .

ANSWER: _____

74. A geophysicist measures the pressure in a borehole. The pressure is 3.975×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

75. Answer the following questions: is I an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

76. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 225 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

77. An engineer measures the capacity of an electricity-generating plant. The capacity is 902 MW. What is the capacity in watts? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

78. Sophia measures the mass of a glass marble using a balance. The accepted mass of the object is 26.7 g. She records a measurement of 27.50 g. What is the percent error of her measurement? Report your answer to two significant figures.

ANSWER: _____

79. Provide the chemical formula for the molecular compound tetraphosphorus decoxide.

ANSWER: _____

80. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 5.12×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

81. Provide the name and empirical formula of the ionic compound that is formed by the Rb^+ cation and the BrO^- anion.

ANSWER: _____

82. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$60 \frac{\text{mol}}{\text{L}} \times 90. \text{L}$$

ANSWER: _____

83. Provide the name of the element that the element symbol U represents.

ANSWER: _____

84. Caleb must note the concentration of a solution of $(\text{H}_3\text{O})_2\text{CO}_3$ in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Caleb weighs an empty graduated cylinder and finds its mass to be 7.9 g. He then puts some solid $(\text{H}_3\text{O})_2\text{CO}_3$ into the graduated cylinder and weighs it. With the $(\text{H}_3\text{O})_2\text{CO}_3$ added, the cylinder weighs 26 g. He adds water to the graduated cylinder and dissolves the $(\text{H}_3\text{O})_2\text{CO}_3$ completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 59.090 mL. What concentration should Caleb write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

85. Write the chemical formula of strontium phosphate decahydrate.

ANSWER: _____

86. A cell biologist measures the width of a cell. The width is $36 \mu\text{m}$. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

87. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.98747 \text{ g} - 0.35 \text{ g}$$

ANSWER: _____

88. Classify the melting point of ice as a physical property or a chemical property. Explain your answer.

ANSWER: _____

89. Benjamin must note the concentration of a solution of NH_4Cl in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Benjamin weighs an empty graduated cylinder and finds its mass to be 5 g. He then puts some solid NH_4Cl into the graduated cylinder and weighs it. With the NH_4Cl added, the cylinder weighs 16.36 g. He adds water to the graduated cylinder and dissolves the NH_4Cl completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 19.1 mL. What concentration should Benjamin write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

90. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 1.00×10^{-14} m. What is the wavelength in femtometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

91. Answer the following questions: is Ba^{2+} an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

92. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.30 \text{ cm} + 53. \text{ cm}$$

ANSWER: _____

93. Does the following process produce a new substance? Vinegar and baking soda are mixed and vigorous bubbling is observed. You must explain your answer with valid reasoning.

ANSWER: _____

94. Identify whether the following describes a random error or a systematic error, and explain your answer: Wind gusts cause repeated outdoor temperature measurements to vary randomly around the true value.

ANSWER: _____

95. John looks up the density of a material and finds it listed as $0.5953 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

96. Write the chemical formula of barium oxide dihydrate.

ANSWER: _____

97. State the SI base unit used to measure time and give its symbol.

ANSWER: _____

98. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 953 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

99. Write the chemical formula of iron (III) oxide heptahydrate.

ANSWER: _____

100. A geologist measures the weight of an ore sample. The weight is 5.9923 kg. What is the weight in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

101. Provide the name and empirical formula of the ionic compound that is formed by the NH_4^+ cation and the O^{2-} anion.

ANSWER: _____

102. Classify the following as an element or a compound: pure nitric acid. You must explain your answer using correct reasoning.

ANSWER: _____

103. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 7.22×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

104. Emily needs to know the mass of a sample of $(\text{NH}_4)_2\text{O}$ to 4 significant figures. She puts the sample on a digital scale. This is what the scale shows: 75.49 g. If this measurement is precise enough for Emily, round it to 4 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

105. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 8.26×10^{10} m. What is the distance in gigameters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

106. A botanist measures the thickness of a leaf. The thickness is 2.6 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

107. A chemist reports a reaction temperature to a U.S. colleague. The reaction temperature is -5°C . Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

108. As part of his duties as an analytical chemist, Dylan measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. He then records these masses in a table in his lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Dylan would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	7.8 g	3.9 g
2	10.7 g	5.3 g
3	11.8 g	3.9 g
4	9.1 g	4.5 g

ANSWER: _____

109. Classify the following as a physical or chemical change: Stretching a rubber band. Explain your answer.

ANSWER: _____

110. A pharmacist measures the mass of a powdered drug. The mass is 3927 mg. What is the mass in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

111. A physical chemist measures the width of a molecule. The width is 20 nm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

112. A millworker measures the width of a plank. The width is 6.0 dm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

113. A geologist measures the weight of an ore sample. The weight is 4.4482 kg. What is the weight in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

114. Provide the name and empirical formula of an ionic compound that is formed by the ammonium cation and the carbonate anion.

ANSWER: _____

115. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$700 \text{ mol} \div 28.1 \text{ L}$$

ANSWER: _____

116. As part of her duties as an analytical chemist, Natalie analyzes a sample of an unknown pure substance X and finds that it contains 6.9 g of element E₁ and 9.3 g of element E₂, and no other elements. She then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
X	6.9 g	9.3 g
A	19.1 g	12.1 g
B	26.2 g	13.6 g
C	8.0 g	19.6 g

ANSWER: _____

117. Identify whether the following describes a random error or a systematic error, and explain your answer: A thermometer is poorly calibrated and reads 2.0 °C too high at every temperature.

ANSWER: _____

118. A pharmacist converts a required storage temperature for a U.S. supplier. The required temperature is 15 °C. Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

119. Provide the name for the molecular compound I₂O₅. If the compound has a common name, you may provide that instead.

ANSWER: _____

120. A geologist measures the temperature of a hot spring. The temperature of the spring is 0.2000°C. Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

121. Joshua must note the concentration of a solution of $\text{Fe}_2(\text{CrO}_4)_3$ in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Joshua weighs an empty graduated cylinder and finds its mass to be 4.10 g. He then puts some solid $\text{Fe}_2(\text{CrO}_4)_3$ into the graduated cylinder and weighs it. With the $\text{Fe}_2(\text{CrO}_4)_3$ added, the cylinder weighs 21 g. He adds water to the graduated cylinder and dissolves the $\text{Fe}_2(\text{CrO}_4)_3$ completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 18.6 mL. What concentration should Joshua write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

122. Provide the number of protons and the number of neutrons there are in an atom of cobalt-59.

ANSWER: _____

123. A measurement is reported in micrograms (μg). How many grams is 1 μg equal to?

ANSWER: _____

124. A recipe instructs a cook to preheat an oven to 50 °F. Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

125. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$15. \text{ mL} + 31. \text{ mL}$$

ANSWER: _____

126. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$8.170 \text{ g} - 0.14 \text{ g}$$

ANSWER: _____

127. State the SI base unit used to measure amount of substance and give its symbol.

ANSWER: _____

128. Does the following process produce a new substance? A copper wire is bent into a coil. You must explain your answer with valid reasoning.

ANSWER: _____

129. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is $4.49 \times 10^{-14} \text{ m}$. What is the wavelength in femtometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

130. Sarah measures the mass of a copper disc using a balance. The accepted mass of the object is 43.8 g. She records a measurement of 43.54 g. What is the percent error of her measurement? Report your answer to two significant figures.

ANSWER: _____

131. Classify the following as a pure substance or a mixture: a sample of liquid mercury in a sealed glass tube. You must explain your answer and include definitions.

ANSWER: _____

132. An optician measures the diameter of her patient's pupil. The diameter is 0.0015 m. What is the diameter in millimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

133. Classify Ag_2O as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

134. Write the chemical formula for nitric acid.

ANSWER: _____

135. Write the chemical formula for arsenic acid.

ANSWER: _____

136. A nurse measures the weight of his patient's medicine. The weight is 0.0054 g. What is the weight in milligrams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

137. Joshua measures the density of an unknown liquid in his lab. He finds that the sample has a mass of 28.48 g and a volume of 57 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

138. Provide the chemical formula for the molecular compound carbon dioxide.

ANSWER: _____

139. A civil engineer measures the length of an airport runway. The length is 6498 m. What is the length in kilometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

140. A nurse records a patient's temperature. The thermometer reads 84.20°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

141. Classify the following as an element or a compound: a lump of pure phosphorus. You must explain your answer using correct reasoning.

ANSWER: _____

142. Provide the chemical symbol of the most commonly formed ion of chlorine and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

143. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 3.00×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

144. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$900\text{mol} \div 31.\text{L}$$

ANSWER: _____

145. A pharmacist measures the mass of a powdered drug. The mass is 2429 mg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

146. Identify whether the following describes a random error or a systematic error, and explain your answer: A student consistently reads a meniscus from above instead of at eye level, always recording a value that is too low.

ANSWER: _____

147. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$41. \frac{\text{mol}}{\text{L}} \times 59.3\text{L}$$

ANSWER: _____

148. Provide the name and empirical formula of an ionic compound that is formed by the caesium cation and the phosphate anion.

ANSWER: _____

149. Provide the element symbol that represents nickel.

ANSWER: _____

150. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.293 m. What is the thickness in centimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

151. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 7.9620×10^{-7} m. What is the wavelength in nanometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

152. A cell biologist measures the width of a cell. The width is 35 μm . What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

153. A geologist measures the height of a mountain. The height is 6.7659 km. What is the height in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

154. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$15. \frac{\text{mol}}{\text{L}} \times 16. \text{L}$$

ANSWER: _____

155. Provide the number of protons and the number of neutrons there are in an atom of plutonium-239.

ANSWER: _____

156. A botanist measures the thickness of a leaf. The thickness is 8.5 mm. What is the thickness in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

157. Provide the name for the molecular compound OF_2 . If the compound has a common name, you may provide that instead.

ANSWER: _____

158. Caleb is comparing two unknown liquid samples. Sample A has a mass of 37.1 g and a volume of 37.2 mL. Sample B has a mass of 26.4 g and a volume of 46.7 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

159. Tyler measures the speed of a moving object and records $25 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

160. Classify particle count as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

161. Paul is a chemical engineer who needs to convert an oven temperature for a thermodynamic calculation. The temperature is -4°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

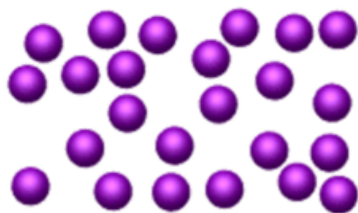
162. A nurse measures the weight of his patient's medicine. The weight is 0.0063 g. What is the weight in milligrams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

163. State the SI base unit used to measure mass and give its symbol.

ANSWER: _____

164. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

165. As part of her duties as an analytical chemist, Grace analyzes a sample of an unknown pure substance X and finds that it contains 11.2 g of element E₁ and 4.9 g of element E₂, and no other elements. She then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
X	11.2 g	4.9 g
A	22.2 g	3.4 g
B	16.9 g	3.7 g
C	36.6 g	3.2 g

ANSWER: _____

166. Provide the chemical symbol of the most commonly formed ion of phosphorus and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

167. A shipwright measures the length of a bolt. The length is 69.2 cm. What is the length in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

168. A lab technician measures the weight of a reagent. The weight is 5966 g. What is the weight in kilograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

169. A geophysicist measures the pressure in a borehole. The pressure is 7.529×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

170. A geologist measures the height of a mountain. The height is 1.3359 km. What is the height in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

171. A nurse measures the weight of his patient's medicine. The weight is 0.0031 g. What is the weight in milligrams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

172. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: a sample of pure sulfur crystals. You must explain your answer with definitions.

ANSWER: _____

173. A pharmacist converts a required storage temperature for a U.S. supplier. The required temperature is 35 °C. Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

174. A millworker measures the width of a plank. The width is 7.5 dm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

175. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 8.20×10^{10} m. What is the distance in gigameters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

176. Allison looks up the density of a material and finds it listed as $9.456 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

177. Nathan needs to know the mass of a sample of $(\text{H}_3\text{O})_2\text{SO}_4$ to 4 significant figures. He puts the sample on a digital scale. This is what the scale shows: 36.49 g. If this measurement is precise enough for Nathan, round it to 4 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

178. Classify the ability to rust of iron as a physical property or a chemical property. Explain your answer.

ANSWER: _____

179. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.391 m. What is the thickness in centimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

180. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 527 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

181. Determine whether potassium and iodine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

182. A geologist measures the height of a mountain. The height is 5.5048 km. What is the height in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

183. A nuclear chemist measures the mass of an artificial element. The mass is 10 pg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

184. What multiplicative factor does the metric prefix 'micro' (symbol: μ) represent?

ANSWER: _____

185. Victoria is comparing two unknown liquid samples. Sample A has a mass of 37 g and a volume of 37.2 mL. Sample B has a mass of 12.3 g and a volume of 6.64 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

186. Answer the following questions: is K^+ an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

187. Emily needs to find the volume of a 19.0 g sample of glycerin. The density of the sample is $4.3 \frac{g}{mL}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

188. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 97 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

189. Provide the name of the element that the element symbol Ga represents.

ANSWER: _____

190. A geophysicist measures the pressure in a borehole. The pressure is 2.128×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

191. Classify temperature as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

192. Identify the state of matter in which oxygen exists at 25 °C and 1 atm.

ANSWER: _____

193. Kyle measures the speed of a moving object and records $49.5 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

194. A nuclear chemist measures the mass of an artificial element. The mass is 78 pg. What is the mass in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

195. As part of her duties as an analytical chemist, Julia measures the amounts of elements E₁ and E₂ found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E₁ and E₂. Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E₁ that Julia would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
1	15.0 g	10.0 g
2	7.0 g	2.6 g
3	10.3 g	10.3 g
4	9.3 g	3.7 g

ANSWER: _____

196. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: a sample of pure sulfur crystals. You must explain your answer with definitions.

ANSWER: _____

197. Jeremy is a chemical engineer who needs to convert an oven temperature for a thermodynamic calculation. The temperature is 41 °F. Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

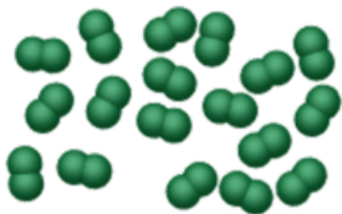
198. A physical chemist measures the width of a molecule. The width is 78 nm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

199. A geologist measures the weight of an ore sample. The weight is 4.6066 kg. What is the weight in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

200. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____